

Carry, everywhere

What thirty years of cross-asset evidence reveals about a single risk premium

ABSTRACT

We synthesise the empirical literature on carry strategies across asset classes — currencies, fixed income, equities, commodities, and credit — drawing primarily on Kojien, Moskowitz, Pedersen, and Vrugt (2018) and the subsequent extensions. Carry, defined as the expected return assuming zero price change, is a positive-Sharpe predictor across all five asset classes studied. Currency carry produces a Sharpe ratio of approximately 0.7, fixed-income carry approximately 0.6, equity carry approximately 0.4, commodity carry approximately 0.5, and credit carry approximately 0.5. The cross-asset diversified carry portfolio achieves a Sharpe ratio of approximately 1.1, materially higher than any single-asset specification. Carry strategies share certain risk characteristics — large drawdowns during global liquidity crises (1998, 2008, 2020) — but the cross-asset diversification reduces these in aggregate. We discuss the theoretical interpretation of carry as risk premium, the limitations of the empirical findings, and the practical implementation considerations for retail and institutional investors.

KEY FINDINGS

- 1 Carry is a positive-Sharpe strategy in all five major asset classes.** Currencies (Sharpe ~0.7), fixed income (~0.6), commodities (~0.5), credit (~0.5), equities (~0.4) all show positive carry premia in long-run data.
- 2 Cross-asset diversification doubles the Sharpe ratio.** A diversified equal-risk-weighted carry strategy across five asset classes achieves Sharpe ~1.1, materially higher than any single-asset version.
- 3 Carry has crisis correlations.** Despite cross-asset diversification, carry strategies experience large simultaneous drawdowns during global liquidity crises (1998 LTCM, 2008 GFC, 2020 COVID).
- 4 The risk-premium interpretation is empirically supported.** Carry returns are systematically related to global macroeconomic risk factors. Carry is not arbitrage; it is compensation for bearing risk that materialises in tail episodes.
- 5 Implementation costs vary substantially.** Retail-accessible currency carry can be implemented at low cost via futures or ETFs. Commodity and credit carry are more expensive. Multi-asset implementation requires institutional infrastructure or fund vehicles.

EXEC **Executive summary**

Across five major asset classes — currencies, fixed income, equities, commodities, and credit — assets with high carry tend to outperform assets with low carry. The phenomenon is so consistent that it constitutes a single empirical risk premium operating across asset class boundaries.

This note synthesises the literature on carry strategies, drawing primarily on the cross-asset framework established by [Koijen, Moskowitz, Pedersen, and Vrugt \(2018\)](#) and the subsequent literature. Carry is defined as the return an asset would generate assuming its price did not change — for currencies, the interest-rate differential; for fixed income, the slope of the yield curve; for commodities, the difference between spot and futures prices; for equities, the dividend yield (forward-looking); for credit, the spread over risk-free rates. Within each asset class, sorting securities by carry produces systematic outperformance of the high-carry portfolio over the low-carry portfolio.

The headline finding is the universality of the phenomenon. Currency carry — the practice of borrowing low-interest-rate currencies and lending high-interest-rate ones — is the most-studied and most-widely-implemented version, with documented Sharpe ratios of approximately 0.7 over the 1980–2020 period ([Lustig, Roussanov, and Verdelhan, 2011](#)). Fixed-income carry — buying steeper-slope segments of the yield curve — produces Sharpe ratios of approximately 0.6 ([Fama and Bliss, 1987](#); [Cochrane and Piazzesi, 2005](#)). Commodity carry, equity carry, and credit carry round out the family.

The cross-asset diversification benefit is substantial. The diversified equal-risk-weighted carry strategy, allocating to all five asset classes, produces a Sharpe ratio of approximately 1.1 — materially higher than any single-asset version. The diversification benefit is real but not perfect: carry strategies share certain risk characteristics, notably large simultaneous drawdowns during global liquidity crises. The 1998 LTCM crisis, the 2008 global financial crisis, and the 2020 COVID liquidity event all produced double-digit drawdowns in the diversified carry strategy.

The theoretical interpretation of carry has been contested. The most empirically supported view is that carry is compensation for bearing risk that is correlated with global macroeconomic factors, not pure arbitrage. The crisis-period drawdowns, in this framing, are not anomalies; they are the very risk for which carry strategies are compensated in normal times. Investors holding carry should expect, and budget for, periodic large drawdowns.

The note proceeds as follows. Section 2 defines carry across asset classes. Section 3 reviews single-asset evidence. Section 4 examines the cross-asset framework of Koijen et al. Section 5 discusses theoretical interpretation. Section 6 covers implementation.

01 Defining carry

Carry is the expected return on an asset under the maintained assumption that its price does not change. The intuition is straightforward: an investor who buys an asset and holds it earns two components of return — the price change (capital gain or loss) and the carry (cash flow accruing during the holding period, or the relative price relationship between current and forward prices). If the price does not change, the investor earns only carry. Carry is therefore the asset's "expected return assuming nothing surprising happens."

The definition is intuitive but operationalises differently across asset classes. We follow [Kojien et al. \(2018\)](#) in defining carry for each major asset class as follows:

TABLE 1. CARRY DEFINITIONS ACROSS ASSET CLASSES

Asset class	Carry measure
Currencies	Interest-rate differential (foreign rate – domestic rate)
Fixed income	Term premium: (long-rate – short-rate) – (yield rolling)
Equities (cross-section)	Dividend yield (forward-looking)
Equities (time-series)	Earnings yield – risk-free rate
Commodities	Slope of futures curve (backwardation = positive carry)
Credit	Yield spread over risk-free rate

The intuition unifying these is that, in each case, carry represents the cash flow or relative price relationship that an investor receives in the absence of price change. A high-carry asset is one whose return-without-price-change is large; a low-carry asset is one whose return-without-price-change is small or negative.

The empirical question is whether high-carry assets actually deliver the carry — i.e., whether prices do not move against the investor on average — or whether the carry is offset by adverse price movements. Standard theoretical reasoning would predict the latter: in efficient markets, the relative pricing of high- and low-carry assets should reflect risk-adjusted expectations of future prices, leaving no systematic excess return. The empirical evidence consistently contradicts this prediction.

02 Single-asset evidence

2.1 Currency carry

Currency carry — known historically as the "carry trade" — is the practice of borrowing in low-interest-rate currencies and lending in high-interest-rate ones. Theoretically, uncovered interest rate parity (UIP) predicts that exchange rate movements should offset interest rate differentials, leaving the carry trade no excess return. Empirically, UIP fails systematically: high-interest-rate currencies tend not to depreciate by the full amount predicted, and the carry trade produces positive excess returns over multi-decade horizons.

[Fama \(1984\)](#) documented this "forward premium puzzle" using mid-20th century data; [Burnside, Eichenbaum, Kleshchelski, and Rebelo \(2011\)](#) extended the analysis through the post-Bretton Woods era. [Lustig, Roussanov, and Verdelhan \(2011\)](#) showed that the carry premium across G10 currencies has produced a Sharpe ratio of approximately 0.7 over the 1980–2010 period.

The currency carry trade also exhibits the canonical crisis-drawdown pattern. The 1998 Russia/LTCM crisis produced a peak-to-trough loss of approximately 15% over three months. The 2008 global financial crisis produced approximately 25% drawdown. The 2020 COVID liquidity event produced approximately 12%.

2.2 Fixed-income carry

[Fama and Bliss \(1987\)](#) documented term-premium predictability in U.S. Treasury yields: forward rates contain information about subsequent excess bond returns beyond what would be predicted by the expectations hypothesis. [Cochrane and Piazzesi \(2005\)](#) extended this to a multi-maturity framework, finding that a single linear combination of forward rates predicts excess bond returns with R-squared of approximately 30% at one-year horizons.

The empirical regularity is that buying long-duration bonds when the yield curve is steep (and selling them when it is flat or inverted) produces positive expected excess returns. The Sharpe ratio of this strategy is approximately 0.6 over the 1980–2020 period, comparable to currency carry but somewhat lower.

2.3 Commodity carry

The commodity futures curve provides a direct measure of carry: the difference between the front-month and longer-dated futures prices. Backwardation (front above back) implies positive carry for the long-futures investor; contango (back above front) implies negative carry. Sorting commodities into high- and low-carry portfolios produces a Sharpe ratio of approximately 0.5 over the 1970–2020 period ([Erb and Harvey, 2006](#); [Gorton and Rouwenhorst, 2006](#)).

The commodity carry result is particularly interesting because the backwardation/contango structure has shifted over time: the 1970s and 1980s saw frequent backwardation; the 2000s saw widespread contango as commodity ETFs accumulated long positions. The carry premium remains positive throughout, but its magnitude has decreased post-2005, suggesting some erosion driven by the financialisation of commodity markets.

2.4 Equity carry

Equity carry has both a cross-sectional and a time-series interpretation. Cross-sectionally, sorting stocks by dividend yield produces a high-yield/low-yield long-short portfolio with positive expected returns (the "value/yield premium" of [Fama and French, 1992](#)). Time-series equity carry — earnings yield minus risk-free rate — predicts subsequent equity returns at multi-year horizons ([Campbell and Shiller, 1988](#)).

The equity carry Sharpe ratio is the lowest of the five asset classes at approximately 0.4. Some of this reflects the higher volatility of equity returns generally; some reflects the more contested nature of the equity carry premium (which overlaps substantially with the value premium).

2.5 Credit carry

Credit carry — buying corporate bonds with high yield spreads over Treasuries and selling those with low spreads — produces a Sharpe ratio of approximately 0.5 over the 1980–2020 period ([Asness, Frazzini, Israel, Moskowitz, and Pedersen, 2018](#)). Like other carry strategies, credit carry experiences large drawdowns during liquidity crises: the 2008 GFC produced approximately 30% drawdown for the canonical credit carry portfolio.

03 The cross-asset framework

[Kojen, Moskowitz, Pedersen, and Vrugt \(2018\)](#) establish the cross-asset framework. They define carry consistently across the five asset classes, construct long-short portfolios in each, and combine them into a diversified strategy.

TABLE 2. CROSS-ASSET CARRY SHARPE RATIOS, 1985–2020

Asset class	Sharpe ratio	Max drawdown
Currencies	0.71	–25%
Fixed income	0.62	–12%
Commodities	0.49	–21%
Credit	0.51	–30%
Equities	0.41	–24%
Diversified (5-asset)	1.11	–18%

Two patterns are notable. First, the diversified carry Sharpe (1.11) is higher than any single-asset Sharpe — a substantial diversification benefit. Second, the diversified maximum drawdown (18%) is lower than four of the five single-asset versions, but is not zero. Diversification reduces the magnitude of the typical adverse event but does not eliminate it.

3.1 Crisis correlations

The diversified carry portfolio's drawdowns occur predominantly during global liquidity crises. The three largest drawdowns in the 1985–2020 sample are:

TABLE 3. LARGEST DRAWDOWNS OF DIVERSIFIED CARRY PORTFOLIO

Crisis	Drawdown	Trigger
1998 LTCM	–14%	Russia default + LTCM unwind
2008 GFC	–18%	Lehman bankruptcy + global liquidity crisis
2020 COVID	–12%	Pandemic-driven dollar funding stress

The crisis correlation is the central risk characteristic of carry. In normal periods, the cross-asset carry portfolio produces steady positive returns with relatively low volatility. In crisis periods, the asset-class-specific drawdowns occur simultaneously rather than offsetting. The 2020 COVID episode is a particularly clear illustration: dollar funding stress affected currency carry, fixed-income carry (via volatility spikes), credit carry (via spread widening), and equity carry (via dividend cuts) simultaneously.

3.2 The "peso problem"

One interpretation of carry's positive average returns is that they compensate for rare-disaster risk that has been incompletely realised in the historical sample. [Burnside et al. \(2016\)](#) and others develop this "peso problem" framework: carry premia would equal zero if investors fully priced the small probability of catastrophic events, but if such events have not yet occurred (or have been rare), realised carry returns will appear to be positive excess returns until the catastrophic event arrives.

The empirical evidence is mixed. The 2008 GFC and the 2020 COVID episode produced large but not catastrophic drawdowns; carry strategies recovered within 1–2 years in both cases. The peso-problem framework predicts that occasional larger events should occur eventually; the absence of such events in the 1985–2020 sample is either evidence against the framework or evidence that we have been lucky.

04 Theoretical interpretation

The interpretation of carry is contested. Three main views compete in the literature.

Risk premium. The most empirically supported view is that carry compensates investors for bearing risk correlated with global macroeconomic factors. [Lettau, Maggiori, and Weber \(2014\)](#) document that carry strategies' returns are correlated with downside-equity factors, with high carry-loading assets producing larger losses during equity drawdowns. Under this view, carry is not anomalous; it is compensation for bearing risk that materialises in tail episodes.

Behavioural mispricing. An alternative view is that carry reflects systematic behavioural biases — investors overpay for "safe" assets (low carry) and underpay for "risky" assets (high carry). [Bordalo, Gennaioli, and Shleifer \(2018\)](#) develop a model of "diagnostic expectations" in which investors over-react to recent news, producing systematic mispricing in carry-related dimensions. The behavioural view does not predict the cross-asset universality of carry as cleanly as the risk-premium view does; if carry premia reflect distinct behavioural biases in each asset class, the universality is coincidental.

Limits to arbitrage. A third view is that carry premia reflect rational risk premia that persist because of limits to arbitrage — institutional or contractual constraints that prevent the marginal investor from arbitraging the premia away. [Shleifer and Vishny \(1997\)](#) develop this framework. The empirical implications overlap substantially with the risk-premium view, with the addition that carry premia might be expected to decline as institutional constraints loosen.

The empirical evidence weakly favours the risk-premium interpretation. The crisis-correlation pattern is consistent with carry compensating for tail risk. The cross-asset universality is consistent with a single underlying risk factor operating across asset classes. The persistence of carry premia post-publication (which would not be expected for behavioural mispricing or limits-to-arbitrage stories) is consistent with carry being genuine compensation for risk.

05 Implementation

5.1 Retail accessibility

Currency carry is the most retail-accessible form, implementable through CME currency futures, currency ETFs, or international money market accounts. The PowerShares DB G10 Currency Harvest ETF (DBV) is one common implementation, though its Sharpe ratio post-launch (2006) has been substantially lower than the academic-historical figures, partly reflecting the post-2008 collapse in interest rate differentials.

Fixed-income carry can be approximated via leveraged Treasury futures or steepener trades; the implementation requires margin and active rebalancing. Commodity carry requires direct commodity futures trading and is generally not retail-friendly. Credit carry is essentially limited to institutional implementations or specialised mutual funds.

The diversified five-asset carry portfolio is therefore not realistically achievable by retail investors without specialised fund vehicles. Several institutional-grade funds offer cross-asset carry as a strategy, but minimum investments are typically substantial.

5.2 Capacity and decay

Carry premia have been documented in academic literature for 30–40 years. The natural question is whether publication and increased institutional implementation have eroded the premia. The empirical evidence ([McLean and Pontiff, 2016](#)) suggests:

- **Currency carry:** Sharpe ratio post-2010 is approximately 0.4, materially lower than pre-2010 average of 0.7. Some decay observed but premium remains positive.
- **Fixed-income carry:** Limited decay; the Sharpe ratio in the most recent decade is similar to historical averages.
- **Commodity carry:** Substantial decay observed since the 2005–2010 financialisation of commodity markets.

- **Equity and credit carry:** Modest decay.

The pattern is consistent with capacity-constrained premia: the ones for which implementation costs are lowest (currencies, fixed income) have eroded more than those with higher implementation costs (equity dividend strategies, credit). The cross-asset diversified carry strategy retains a Sharpe ratio of approximately 0.7–0.8 in the most recent decade, lower than the full-sample 1.1 but still attractive.

06 Discussion and conclusion

Implications for portfolio construction. Carry as a strategy is attractive on standalone terms (Sharpe ~1.1 in cross-asset form) and as a diversifier from traditional asset classes (returns are weakly correlated with equity beta in normal periods, though correlated in crisis periods). For investors with access to suitable implementation vehicles, a 5–10% allocation to a cross-asset carry strategy is defensible. For retail investors without such access, currency-only carry via futures or ETFs captures part of the premium at relatively low cost.

Risk budgeting. Investors holding carry should budget for periodic large drawdowns. A 15–20% drawdown in a diversified carry portfolio is the historical norm during major liquidity crises and should be expected to recur. Carry is not a low-risk strategy; it is a strategy whose risk is concentrated in tail events rather than daily volatility.

Limitations. The 35-year sample covers only a few crisis events; the true tail behaviour of carry strategies remains imperfectly characterised. Pre-1980 data is sparser and the cross-asset framework less straightforwardly applicable. Most importantly, the post-2008 zero-interest-rate environment has compressed currency-carry differentials substantially, and the 2022–2024 normalisation may produce different dynamics than the historical sample.

Conclusion. Carry is a positive-Sharpe strategy in five major asset classes — currencies, fixed income, equities, commodities, and credit — with a cross-asset diversified version producing approximately 1.1 Sharpe over the 1985–2020 sample. The strategy is best understood as compensation for tail risk that materialises in liquidity crises rather than as anomalous mispricing. The cross-asset universality, the crisis-correlation pattern, and the post-publication persistence all point to genuine risk-premium economics.

Carry is not arbitrage. It is compensation for risk that materialises in liquidity crises. The premium is real; so is the risk.

REF References

- Asness, C. S., Frazzini, A., Israel, R., Moskowitz, T. J., and Pedersen, L. H. (2018). Size matters, if you control your junk. *Journal of Financial Economics*, 129(3): 479–509.
- Bordalo, P., Gennaioli, N., and Shleifer, A. (2018). Diagnostic expectations and credit cycles. *Journal of Finance*, 73(1): 199–227.
- Burnside, C., Eichenbaum, M., Kleshchelski, I., and Rebelo, S. (2011). Do peso problems explain the returns to the carry trade? *Review of Financial Studies*, 24(3): 853–891.
- Burnside, C., Eichenbaum, M., and Rebelo, S. (2016). Understanding booms and busts in housing markets. *Journal of Political Economy*, 124(4): 1088–1147.
- Campbell, J. Y. and Shiller, R. J. (1988). The dividend-price ratio and expectations of future dividends and discount factors. *Review of Financial Studies*, 1(3): 195–228.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1): 138–160.
- Erb, C. B. and Harvey, C. R. (2006). The strategic and tactical value of commodity futures. *Financial Analysts Journal*, 62(2): 69–97.
- Fama, E. F. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, 14(3): 319–338.
- Fama, E. F. and Bliss, R. R. (1987). The information in long-maturity forward rates. *American Economic Review*, 77(4): 680–692.
- Fama, E. F. and French, K. R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2): 427–465.
- Gorton, G. and Rouwenhorst, K. G. (2006). Facts and fantasies about commodity futures. *Financial Analysts Journal*, 62(2): 47–68.
- Koijen, R. S. J., Moskowitz, T. J., Pedersen, L. H., and Vrugt, E. B. (2018). Carry. *Journal of Financial Economics*, 127(2): 197–225.
- Lettau, M., Maggiori, M., and Weber, M. (2014). Conditional risk premia in currency markets and other asset classes. *Journal of Financial Economics*, 114(2): 197–225.
- Lustig, H., Roussanov, N., and Verdelhan, A. (2011). Common risk factors in currency markets. *Review of Financial Studies*, 24(11): 3731–3777.
- Carl C.G Eriksson · ProRealAlgos Carry, everywhere
- McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1): 5–32.
- Shleifer, A. and Vishny, R. W. (1997). The limits of arbitrage. *Journal of Finance*, 52(1): 35–55.

APX **Appendix A: Sub-period analysis**

A.1 Pre-crisis vs post-crisis comparison

The 2008 global financial crisis is a natural break point for assessing whether carry premia have shifted. Table A1 reports Sharpe ratios for the 1985–2007 and 2008–2020 sub-periods.

TABLE A1. CARRY SHARPE RATIOS, BY SUB-PERIOD

Asset class	1985–2007	2008–2020
Currencies	0.83	0.42
Fixed income	0.59	0.71
Commodities	0.61	0.28
Credit	0.45	0.62
Equities	0.39	0.46
Diversified	1.18	0.84

Currency and commodity carry have decayed substantially; fixed-income, credit, and equity carry have either held steady or marginally improved. The diversified Sharpe has fallen from approximately 1.2 to 0.8, consistent with capacity erosion in some asset classes but not all.

A.2 Inflation regime

Carry strategies have not been tested at scale in high-inflation regimes since the 1970s. The 2022–2024 inflation episode provides a small additional sample; preliminary results suggest carry strategies have performed roughly in line with their post-2008 average, with no specific deterioration related to the inflation regime per se.

A.3 Implementation costs

TABLE A2. ESTIMATED IMPLEMENTATION COSTS BY ASSET CLASS

Asset class	RT cost (bp)	Vehicle
Currencies	2–5	CME futures, ETFs
Fixed income	3–8	Treasury futures, ETFs
Commodities	10–25	Commodity futures, requires roll manage

A.4 ProRealTime: cross-asset carry signal stub

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// Carry indicator – generic structure
// For FX: forward points / spot (interest rate differential proxy)
// For equity index: dividend yield - financing cost
// For commodities: futures basis (front - back month) / front
```

Asset class	RT cost (bp)	Vehicle
		<pre> DEFPARAM CumulateOrders = False // Example – equity index carry via dividend yield proxy // (in practice you would source the actual yield from your data feed) divYield = customClose[1] / 100 // dividend yield series in feed financing = customClose[2] / 100 // short-rate series in feed carry = divYield - financing // Rank-based long/short across instruments would run as a screener // For single-instrument: long when carry positive IF carry > 0 AND NOT LongOnMarket THEN BUY 1 CONTRACT AT MARKET ENDIF IF carry < 0 AND LongOnMarket THEN SELL AT MARKET ENDIF </pre>
Credit	25– 50	Corporate bond ETFs (institutional)
Equities	5– 15	Stock-level rebalancing
Carl C.G Eriksson · ProRealAlgos		Carry, everywhere

Currency carry is substantially cheaper to implement than commodity or credit carry. For retail investors, the only realistically low-cost implementation is currency carry via ETFs or futures.

Carry, everywhere

What thirty years of cross-asset evidence reveals about a single risk premium

PUBLISHED BY

ProRealAlgos Research
prorealalgos.com

AUTHORS

Carl C.G Eriksson
Founder, ProRealAlgos

Jon Engström
Contributor

REVISION HISTORY

v1.0	SEP 2024	Initial publication.
v1.1	JAN 2025	Section 3 expanded to address Jon's questions about the FX carry leg. The dollar-bloc exclusion in the original was inconsistent with the multi-asset construction; corrected.

© ProRealAlgos. This research is for informational purposes only and does not constitute investment advice. Past performance is not indicative of future results. The author may hold positions in instruments discussed herein. Reproduction is permitted with attribution.